

Workforce Data Analysis: Salary and Remote Work Trends Across Job Titles and Company Locations (CaseStudy)

1. Introduction

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Organizations increasingly rely on workforce analytics to make informed decisions regarding employee compensation, recruitment, remote work policies, and talent retention. This project analyzes a workforce salary dataset containing information about job titles, experience levels, company sizes, employee locations, salaries, and remote work arrangements.

The objective of this study is to identify salary trends, compare compensation across countries and experience levels, evaluate remote work adoption, and generate insights that can help organizations improve workforce planning and strategic decision-making.

The analysis was performed using SQL Server and includes various business-oriented queries designed to uncover patterns within the workforce dataset.

2. Dataset Description

The dataset contains **13,972 records** and **11 attributes** related to employee salaries and employment information.

Attributes Used

Column Name	Description
work_year	Year of employment
experience_level	Employee experience level
employment_type	Full-Time or Part-Time
job_title	Employee designation
salary	Original salary
salary_currency	Salary currency
salary_in_usd	Salary converted to USD
employee_residence	Employee country
remote_ratio	Remote work percentage
company_location	Company country
company_size	Small, Medium, Large

Dataset period: **2020–2024**.

Key experience levels:

- EN – Entry Level
- MI – Mid Level
- SE – Senior Level
- EX – Executive Level

3. Objectives

The major objectives of the project are:

- Analyze salary trends across countries.
- Compare salaries by experience level.
- Study remote work adoption.
- Identify high-paying job titles.
- Evaluate company size distribution.
- Discover workforce growth opportunities.
- Support strategic business decisions.
- Improve talent acquisition and retention strategies.

4. Methodology

Step 1: Data Collection

The workforce salary dataset was imported into SQL Server.

Step 2: Data Understanding

Dataset fields were examined and validated.

Step 3: Data Cleaning

- Removed inconsistencies
- Verified null values
- Standardized salary fields

Step 4: SQL Analysis

Business questions were solved using:

- SELECT
- WHERE
- GROUP BY
- HAVING
- ORDER BY
- CASE
- UPDATE
- Aggregate Functions
- Subqueries

Step 5: Insight Generation

Results were analyzed to identify workforce trends and salary patterns.

5. Key Findings

Workforce Distribution

- Senior-level employees represent the largest portion of the workforce.
- Mid-level employees form the second-largest workforce group.
- Executive-level employees represent the smallest segment.

Popular Job Roles

Top job roles include:

1. Data Engineer

2. Data Scientist
3. Data Analyst
4. Machine Learning Engineer
5. Research Scientist

Salary Trends

- Executive and senior-level employees earn significantly higher salaries.
- Salary increases consistently with experience level.
- Specialized AI and Machine Learning roles attract premium salaries.

Company Distribution

- Large companies employ a significant portion of the workforce.
- The United States hosts the majority of organizations in the dataset.

Remote Work Analysis

- Most employees still work in-office.
- Fully remote jobs represent a significant portion of the workforce.
- Hybrid roles are comparatively limited.

6. Business Insights

Salary Benchmarking

Organizations can use salary data to establish competitive compensation structures and reduce employee turnover.

Talent Acquisition

Countries offering competitive salaries can be targeted for recruitment and expansion activities.

Workforce Planning

Understanding experience-level distributions helps companies design effective hiring strategies.

Remote Work Optimization

Remote work trends can help organizations design flexible workplace policies.

Cost Optimization

Companies can strategically hire talent in regions where salaries are competitive while maintaining productivity.

7. Actionable Recommendations

Recommendation 1

Implement market-based salary benchmarking to remain competitive.

Recommendation 2

Expand hiring in countries showing strong salary growth.

Recommendation 3

Increase remote work opportunities for high-demand roles.

Recommendation 4

Develop retention strategies for senior-level professionals.

Recommendation 5

Invest in training programs that help employees transition into high-paying AI and Data Science roles.

8. Conclusion

The Workforce Data Analysis project successfully demonstrated the use of SQL for workforce and salary analytics. The analysis revealed significant differences in salaries across experience levels, job titles, company locations, and remote work arrangements.

The study highlighted the importance of salary benchmarking, workforce planning, remote work adoption, and talent retention strategies. Organizations can leverage these insights to improve hiring decisions, optimize compensation structures, and remain competitive in the evolving job market.

Overall, the project provided valuable insights into global workforce trends and demonstrated how data-driven decision-making can enhance business performance and employee satisfaction.